

Trajectory_Analysis_Perturbations_PBMC

June 1, 2026

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[ ]: !pip install scanpy
      !pip install anndata
      !pip3 install igraph
      !pip install celltypist
      !pip install decoupler
      !pip install fa2-modified
      !pip install louvain
      !pip install perpy
```

Collecting scanpy

Downloading scanpy-1.12-py3-none-any.whl.metadata (8.4 kB)

Collecting anndata>=0.10.8 (from scanpy)

Downloading anndata-0.12.10-py3-none-any.whl.metadata (9.9 kB)

Collecting fast-array-utils>=1.2.1 (from fast-array-utils[accel,sparse]>=1.2.1->scanpy)

Downloading fast_array_utils-1.3.1-py3-none-any.whl.metadata (3.9 kB)

Requirement already satisfied: h5py>=3.11 in /usr/local/lib/python3.12/dist-packages (from scanpy) (3.15.1)

Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages (from scanpy) (1.5.3)

Collecting legacy-api-wrap>=1.5 (from scanpy)

Downloading legacy_api_wrap-1.5-py3-none-any.whl.metadata (2.2 kB)

Requirement already satisfied: matplotlib>=3.9 in /usr/local/lib/python3.12/dist-packages (from scanpy) (3.10.0)

Requirement already satisfied: natsort in /usr/local/lib/python3.12/dist-packages (from scanpy) (8.4.0)

Requirement already satisfied: networkx>=2.8.8 in /usr/local/lib/python3.12/dist-packages (from scanpy) (3.6.1)

Requirement already satisfied: numba>=0.60 in /usr/local/lib/python3.12/dist-packages (from scanpy) (0.60.0)

Requirement already satisfied: numpy>=2 in /usr/local/lib/python3.12/dist-packages (from scanpy) (2.0.2)

Requirement already satisfied: packaging>=25 in /usr/local/lib/python3.12/dist-packages (from scanpy) (26.0)

Requirement already satisfied: pandas>=2.2.2 in /usr/local/lib/python3.12/dist-packages (from scanpy) (2.2.2)

Requirement already satisfied: patsy in /usr/local/lib/python3.12/dist-packages (from scanpy) (1.0.2)

Requirement already satisfied: pynndescent>=0.5.13 in /usr/local/lib/python3.12/dist-packages (from scanpy) (0.6.0)

Requirement already satisfied: scikit-learn>=1.4.2 in /usr/local/lib/python3.12/dist-packages (from scanpy) (1.6.1)

Requirement already satisfied: scipy>=1.13 in /usr/local/lib/python3.12/dist-packages (from scanpy) (1.16.3)

Requirement already satisfied: seaborn>=0.13.2 in /usr/local/lib/python3.12/dist-packages (from scanpy) (0.13.2)

Collecting session-info2 (from scanpy)

 Downloading session_info2-0.4-py3-none-any.whl.metadata (3.5 kB)

Requirement already satisfied: statsmodels>=0.14.5 in /usr/local/lib/python3.12/dist-packages (from scanpy) (0.14.6)

Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from scanpy) (4.67.2)

Requirement already satisfied: typing-extensions in /usr/local/lib/python3.12/dist-packages (from scanpy) (4.15.0)

Requirement already satisfied: umap-learn>=0.5.7 in /usr/local/lib/python3.12/dist-packages (from scanpy) (0.5.11)

Collecting array-api-compat>=1.7.1 (from anndata>=0.10.8->scanpy)

 Downloading array_api_compat-1.13.0-py3-none-any.whl.metadata (2.5 kB)

Collecting zarr!=3.0.*,>=2.18.7 (from anndata>=0.10.8->scanpy)

 Downloading zarr-3.1.5-py3-none-any.whl.metadata (10 kB)

Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.9->scanpy) (1.3.3)

Requirement already satisfied: cycycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.9->scanpy) (0.12.1)

Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.9->scanpy) (4.61.1)

Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.9->scanpy) (1.4.9)

Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.9->scanpy) (11.3.0)

Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.9->scanpy) (3.3.2)

Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.9->scanpy) (2.9.0.post0)

Requirement already satisfied: llvmlite<0.44,>=0.43.0dev0 in /usr/local/lib/python3.12/dist-packages (from numba>=0.60->scanpy) (0.43.0)

Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas>=2.2.2->scanpy) (2025.2)

Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas>=2.2.2->scanpy) (2025.3)

Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn>=1.4.2->scanpy) (3.6.0)

Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib>=3.9->scanpy) (1.17.0)

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Collecting donfig>=0.8 (from zarr!=3.0.*,>=2.18.7->anndata>=0.10.8->scanpy)
  Downloading donfig-0.8.1.post1-py3-none-any.whl.metadata (5.0 kB)
Requirement already satisfied: google-crc32c>=1.5 in
/usr/local/lib/python3.12/dist-packages (from
zarr!=3.0.*,>=2.18.7->anndata>=0.10.8->scanpy) (1.8.0)
Collecting numcodecs>=0.14 (from zarr!=3.0.*,>=2.18.7->anndata>=0.10.8->scanpy)
  Downloading numcodecs-0.16.5-cp312-cp312-
manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata
(3.4 kB)
Requirement already satisfied: pyyaml in /usr/local/lib/python3.12/dist-packages
(from donfig>=0.8->zarr!=3.0.*,>=2.18.7->anndata>=0.10.8->scanpy) (6.0.3)
Downloading scanpy-1.12-py3-none-any.whl (2.1 MB)
      2.1/2.1 MB
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Downloading anndata-0.12.10-py3-none-any.whl (176 kB)
      176.6/176.6 kB
16.0 MB/s eta 0:00:00
Downloading fast_array_utils-1.3.1-py3-none-any.whl (36 kB)
Downloading legacy_api_wrap-1.5-py3-none-any.whl (10 kB)
Downloading session_info2-0.4-py3-none-any.whl (17 kB)
Downloading array_api_compat-1.13.0-py3-none-any.whl (58 kB)
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Downloading zarr-3.1.5-py3-none-any.whl (284 kB)
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16.5 MB/s eta 0:00:00
Downloading donfig-0.8.1.post1-py3-none-any.whl (21 kB)
Downloading numcodecs-0.16.5-cp312-cp312-
manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (9.2 MB)
      9.2/9.2 MB
79.6 MB/s eta 0:00:00
Installing collected packages: session-info2, numcodecs, legacy-api-wrap,
fast-array-utils, donfig, array-api-compat, zarr, anndata, scanpy
Successfully installed anndata-0.12.10 array-api-compat-1.13.0
donfig-0.8.1.post1 fast-array-utils-1.3.1 legacy-api-wrap-1.5 numcodecs-0.16.5
scanpy-1.12 session-info2-0.4 zarr-3.1.5
Requirement already satisfied: anndata in /usr/local/lib/python3.12/dist-
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Requirement already satisfied: array-api-compat>=1.7.1 in
/usr/local/lib/python3.12/dist-packages (from anndata) (1.13.0)
Requirement already satisfied: h5py>=3.8 in /usr/local/lib/python3.12/dist-
packages (from anndata) (3.15.1)
Requirement already satisfied: legacy-api-wrap in
/usr/local/lib/python3.12/dist-packages (from anndata) (1.5)
Requirement already satisfied: natsort in /usr/local/lib/python3.12/dist-
packages (from anndata) (8.4.0)
Requirement already satisfied: numpy>=1.26 in /usr/local/lib/python3.12/dist-
packages (from anndata) (2.0.2)

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Requirement already satisfied: packaging>=24.2 in /usr/local/lib/python3.12/dist-packages (from anndata) (26.0)

Requirement already satisfied: pandas!=2.1.2,<3,>=2.1.0 in /usr/local/lib/python3.12/dist-packages (from anndata) (2.2.2)

Requirement already satisfied: scipy!=1.17.0,>=1.12 in /usr/local/lib/python3.12/dist-packages (from anndata) (1.16.3)

Requirement already satisfied: zarr!=3.0.*,>=2.18.7 in /usr/local/lib/python3.12/dist-packages (from anndata) (3.1.5)

Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas!=2.1.2,<3,>=2.1.0->anndata) (2.9.0.post0)

Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas!=2.1.2,<3,>=2.1.0->anndata) (2025.2)

Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas!=2.1.2,<3,>=2.1.0->anndata) (2025.3)

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Requirement already satisfied: google-crc32c>=1.5 in /usr/local/lib/python3.12/dist-packages (from zarr!=3.0.*,>=2.18.7->anndata) (1.8.0)

Requirement already satisfied: numcodecs>=0.14 in /usr/local/lib/python3.12/dist-packages (from zarr!=3.0.*,>=2.18.7->anndata) (0.16.5)

Requirement already satisfied: typing-extensions>=4.9 in /usr/local/lib/python3.12/dist-packages (from zarr!=3.0.*,>=2.18.7->anndata) (4.15.0)

Requirement already satisfied: pyyaml in /usr/local/lib/python3.12/dist-packages (from donfig>=0.8->zarr!=3.0.*,>=2.18.7->anndata) (6.0.3)

Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas!=2.1.2,<3,>=2.1.0->anndata) (1.17.0)

Collecting igraph

 Downloading igraph-1.0.0-cp39-abi3-manylinux_2_28_x86_64.whl.metadata (4.4 kB)

Collecting texttable>=1.6.2 (from igraph)

 Downloading texttable-1.7.0-py2.py3-none-any.whl.metadata (9.8 kB)

Downloading igraph-1.0.0-cp39-abi3-manylinux_2_28_x86_64.whl (5.7 MB)

5.7/5.7 MB

68.7 MB/s eta 0:00:00

Downloading texttable-1.7.0-py2.py3-none-any.whl (10 kB)

Installing collected packages: texttable, igraph

Successfully installed igraph-1.0.0 texttable-1.7.0

Collecting celltypist

 Downloading celltypist-1.7.1-py3-none-any.whl.metadata (45 kB)

45.1/45.1 kB

3.0 MB/s eta 0:00:00

Requirement already satisfied: numpy>=1.19.0 in /usr/local/lib/python3.12/dist-packages (from celltypist) (2.0.2)

Requirement already satisfied: pandas>=1.0.5 in /usr/local/lib/python3.12/dist-

packages (from celltypist) (2.2.2)
 Requirement already satisfied: scikit-learn>=0.24.1 in
 /usr/local/lib/python3.12/dist-packages (from celltypist) (1.6.1)
 Requirement already satisfied: openpyxl>=3.0.4 in
 /usr/local/lib/python3.12/dist-packages (from celltypist) (3.1.5)
 Requirement already satisfied: click>=7.1.2 in /usr/local/lib/python3.12/dist-
 packages (from celltypist) (8.3.1)
 Requirement already satisfied: requests>=2.23.0 in
 /usr/local/lib/python3.12/dist-packages (from celltypist) (2.32.4)
 Requirement already satisfied: scanpy>=1.7.0 in /usr/local/lib/python3.12/dist-
 packages (from celltypist) (1.12)
 Collecting leidenalg>=0.9.0 (from celltypist)
 Downloading leidenalg-0.11.0-cp38-abi3-
 manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (10 kB)
 Requirement already satisfied: igraph<2.0,>=1.0.0 in
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 Requirement already satisfied: et-xmlfile in /usr/local/lib/python3.12/dist-
 packages (from openpyxl>=3.0.4->celltypist) (2.0.0)
 Requirement already satisfied: python-dateutil>=2.8.2 in
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 Requirement already satisfied: charset_normalizer<4,>=2 in
 /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->celltypist)
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 Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-
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 Requirement already satisfied: urllib3<3,>=1.21.1 in
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 Requirement already satisfied: certifi>=2017.4.17 in
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 Requirement already satisfied: anndata>=0.10.8 in
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 Requirement already satisfied: fast-array-utils>=1.2.1 in
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 utils[accel,sparse]>=1.2.1->scanpy>=1.7.0->celltypist) (1.3.1)
 Requirement already satisfied: h5py>=3.11 in /usr/local/lib/python3.12/dist-
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 Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages
 (from scanpy>=1.7.0->celltypist) (1.5.3)
 Requirement already satisfied: legacy-api-wrap>=1.5 in

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 Requirement already satisfied: matplotlib>=3.9 in
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 Requirement already satisfied: natsort in /usr/local/lib/python3.12/dist-
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 Requirement already satisfied: networkx>=2.8.8 in
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 Requirement already satisfied: numba>=0.60 in /usr/local/lib/python3.12/dist-
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 Requirement already satisfied: patsy in /usr/local/lib/python3.12/dist-packages
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 Requirement already satisfied: pynndescent>=0.5.13 in
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 Requirement already satisfied: scipy>=1.13 in /usr/local/lib/python3.12/dist-
 packages (from scanpy>=1.7.0->celltypist) (1.16.3)
 Requirement already satisfied: seaborn>=0.13.2 in
 /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celltypist)
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 Requirement already satisfied: statsmodels>=0.14.5 in
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 Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages
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 Requirement already satisfied: threadpoolctl>=3.1.0 in
 /usr/local/lib/python3.12/dist-packages (from scikit-learn>=0.24.1->celltypist)
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 Requirement already satisfied: zarr!=3.0.*,>=2.18.7 in
 /usr/local/lib/python3.12/dist-packages (from
 anndata>=0.10.8->scanpy>=1.7.0->celltypist) (3.1.5)
 Requirement already satisfied: texttable>=1.6.2 in
 /usr/local/lib/python3.12/dist-packages (from
 igraph<2.0,>=1.0.0->leidenalg>=0.9.0->celltypist) (1.7.0)
 Requirement already satisfied: contourpy>=1.0.1 in
 /usr/local/lib/python3.12/dist-packages (from

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matplotlib>=3.9->scanpy>=1.7.0->celltypist) (1.3.3)
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matplotlib>=3.9->scanpy>=1.7.0->celltypist) (4.61.1)
Requirement already satisfied: kiwisolver>=1.3.1 in
/usr/local/lib/python3.12/dist-packages (from
matplotlib>=3.9->scanpy>=1.7.0->celltypist) (1.4.9)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-
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Requirement already satisfied: pyparsing>=2.3.1 in
/usr/local/lib/python3.12/dist-packages (from
matplotlib>=3.9->scanpy>=1.7.0->celltypist) (3.3.2)
Requirement already satisfied: llvmlite<0.44,>=0.43.0dev0 in
/usr/local/lib/python3.12/dist-packages (from
numba>=0.60->scanpy>=1.7.0->celltypist) (0.43.0)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-
packages (from python-dateutil>=2.8.2->pandas>=1.0.5->celltypist) (1.17.0)
Requirement already satisfied: donfig>=0.8 in /usr/local/lib/python3.12/dist-
packages (from zarr!=3.0.*,>=2.18.7->anndata>=0.10.8->scanpy>=1.7.0->celltypist)
(0.8.1.post1)
Requirement already satisfied: google-crc32c>=1.5 in
/usr/local/lib/python3.12/dist-packages (from
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/usr/local/lib/python3.12/dist-packages (from
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Requirement already satisfied: pyyaml in /usr/local/lib/python3.12/dist-packages
(from
donfig>=0.8->zarr!=3.0.*,>=2.18.7->anndata>=0.10.8->scanpy>=1.7.0->celltypist)
(6.0.3)
Downloading celltypist-1.7.1-py3-none-any.whl (7.3 MB)
7.3/7.3 MB
28.2 MB/s eta 0:00:00
Downloading
leidenalg-0.11.0-cp38-abi3-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (2.7
MB)
2.7/2.7 MB
70.6 MB/s eta 0:00:00
Installing collected packages: leidenalg, celltypist
Successfully installed celltypist-1.7.1 leidenalg-0.11.0
Collecting decoupler
  Downloading decoupler-2.1.4-py3-none-any.whl.metadata (8.8 kB)
Collecting adjusttext (from decoupler)
  Downloading adjustText-1.3.0-py3-none-any.whl.metadata (3.1 kB)
Requirement already satisfied: anndata in /usr/local/lib/python3.12/dist-
packages (from decoupler) (0.12.10)

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Collecting docrep (from decoupler)
  Downloading docrep-0.3.2.tar.gz (33 kB)
  Preparing metadata (setup.py) ... done
Collecting marsilea (from decoupler)
  Downloading marsilea-0.5.8-py3-none-any.whl.metadata (7.3 kB)
Requirement already satisfied: numba in /usr/local/lib/python3.12/dist-packages
(from decoupler) (0.60.0)
Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-
packages (from decoupler) (2.32.4)
Requirement already satisfied: scipy in /usr/local/lib/python3.12/dist-packages
(from decoupler) (1.16.3)
Requirement already satisfied: session-info2 in /usr/local/lib/python3.12/dist-
packages (from decoupler) (0.4)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages
(from decoupler) (4.67.2)
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages
(from adjusttext->decoupler) (2.0.2)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-
packages (from adjusttext->decoupler) (3.10.0)
Requirement already satisfied: array-api-compat>=1.7.1 in
/usr/local/lib/python3.12/dist-packages (from anndata->decoupler) (1.13.0)
Requirement already satisfied: h5py>=3.8 in /usr/local/lib/python3.12/dist-
packages (from anndata->decoupler) (3.15.1)
Requirement already satisfied: legacy-api-wrap in
/usr/local/lib/python3.12/dist-packages (from anndata->decoupler) (1.5)
Requirement already satisfied: natsort in /usr/local/lib/python3.12/dist-
packages (from anndata->decoupler) (8.4.0)
Requirement already satisfied: packaging>=24.2 in
/usr/local/lib/python3.12/dist-packages (from anndata->decoupler) (26.0)
Requirement already satisfied: pandas!=2.1.2,<3,>=2.1.0 in
/usr/local/lib/python3.12/dist-packages (from anndata->decoupler) (2.2.2)
Requirement already satisfied: zarr!=3.0.*,>=2.18.7 in
/usr/local/lib/python3.12/dist-packages (from anndata->decoupler) (3.1.5)
Requirement already satisfied: six in /usr/local/lib/python3.12/dist-packages
(from docrep->decoupler) (1.17.0)
Collecting legendkit (from marsilea->decoupler)
  Downloading legendkit-0.3.6-py3-none-any.whl.metadata (2.6 kB)
Requirement already satisfied: platformdirs in /usr/local/lib/python3.12/dist-
packages (from marsilea->decoupler) (4.5.1)
Requirement already satisfied: seaborn in /usr/local/lib/python3.12/dist-
packages (from marsilea->decoupler) (0.13.2)
Requirement already satisfied: llvmlite<0.44,>=0.43.0dev0 in
/usr/local/lib/python3.12/dist-packages (from numba->decoupler) (0.43.0)
Requirement already satisfied: charset_normalizer<4,>=2 in
/usr/local/lib/python3.12/dist-packages (from requests->decoupler) (3.4.4)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-
packages (from requests->decoupler) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in

```



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/usr/local/lib/python3.12/dist-packages (from requests->decoupler) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in
/usr/local/lib/python3.12/dist-packages (from requests->decoupler) (2026.1.4)
Requirement already satisfied: contourpy>=1.0.1 in
/usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->decoupler)
(1.3.3)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-
packages (from matplotlib->adjusttext->decoupler) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in
/usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->decoupler)
(4.61.1)
Requirement already satisfied: kiwisolver>=1.3.1 in
/usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->decoupler)
(1.4.9)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-
packages (from matplotlib->adjusttext->decoupler) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in
/usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->decoupler)
(3.3.2)
Requirement already satisfied: python-dateutil>=2.7 in
/usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->decoupler)
(2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-
packages (from pandas!=2.1.2,<3,>=2.1.0->anndata->decoupler) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-
packages (from pandas!=2.1.2,<3,>=2.1.0->anndata->decoupler) (2025.3)
Requirement already satisfied: donfig>=0.8 in /usr/local/lib/python3.12/dist-
packages (from zarr!=3.0.*,>=2.18.7->anndata->decoupler) (0.8.1.post1)
Requirement already satisfied: google-crc32c>=1.5 in
/usr/local/lib/python3.12/dist-packages (from
zarr!=3.0.*,>=2.18.7->anndata->decoupler) (1.8.0)
Requirement already satisfied: numcodecs>=0.14 in
/usr/local/lib/python3.12/dist-packages (from
zarr!=3.0.*,>=2.18.7->anndata->decoupler) (0.16.5)
Requirement already satisfied: typing-extensions>=4.9 in
/usr/local/lib/python3.12/dist-packages (from
zarr!=3.0.*,>=2.18.7->anndata->decoupler) (4.15.0)
Requirement already satisfied: pyarrow>=10.0.1 in
/usr/local/lib/python3.12/dist-packages (from
pandas[parquet]->marsilea->decoupler) (18.1.0)
Requirement already satisfied: pyyaml in /usr/local/lib/python3.12/dist-packages
(from donfig>=0.8->zarr!=3.0.*,>=2.18.7->anndata->decoupler) (6.0.3)
Downloading decoupler-2.1.4-py3-none-any.whl (121 kB)
121.8/121.8 kB
1.3 MB/s eta 0:00:00
Downloading adjustText-1.3.0-py3-none-any.whl (13 kB)
Downloading marsilea-0.5.8-py3-none-any.whl (84 kB)
85.0/85.0 kB

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6.4 MB/s eta 0:00:00

Downloading legendkit-0.3.6-py3-none-any.whl (24 kB)

Building wheels for collected packages: docrep

Building wheel for docrep (setup.py) ... done

Created wheel for docrep: filename=docrep-0.3.2-py3-none-any.whl size=19876
sha256=a522ea4656e2c4ec1e15ec79dda1fd2b160a6ecf6f3e4cfb43d2e10b917bfacd

Stored in directory: /root/.cache/pip/wheels/d6/19/ee/0a6a1793d91c449563b49cca
b57ce52da3e6fab7614836bd8c

Successfully built docrep

Installing collected packages: docrep, legendkit, adjusttext, marsilea,
decoupler

Successfully installed adjusttext-1.3.0 decoupler-2.1.4 docrep-0.3.2

legendkit-0.3.6 marsilea-0.5.8

Collecting fa2-modified

Downloading fa2_modified-0.4-cp312-cp312-
manylinux1_x86_64.manylinux_2_28_x86_64.manylinux_2_5_x86_64.whl.metadata (9.3
kB)

Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages
(from fa2-modified) (2.0.2)

Requirement already satisfied: scipy in /usr/local/lib/python3.12/dist-packages
(from fa2-modified) (1.16.3)

Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages
(from fa2-modified) (4.67.2)

Downloading fa2_modified-0.4-cp312-cp312-
manylinux1_x86_64.manylinux_2_28_x86_64.manylinux_2_5_x86_64.whl (620 kB)

620.8/620.8 kB

14.1 MB/s eta 0:00:00

Installing collected packages: fa2-modified

Successfully installed fa2-modified-0.4

Collecting louvain

Downloading louvain-0.8.2.tar.gz (4.2 MB)

4.2/4.2 MB

58.2 MB/s eta 0:00:00

Preparing metadata (setup.py) ... done

Collecting igraph<0.12,>=0.10.0 (from louvain)

Downloading

igraph-0.11.9-cp39-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata
(4.4 kB)

Requirement already satisfied: texttable>=1.6.2 in
/usr/local/lib/python3.12/dist-packages (from igraph<0.12,>=0.10.0->louvain)
(1.7.0)

Downloading

igraph-0.11.9-cp39-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (4.4 MB)

4.4/4.4 MB

70.9 MB/s eta 0:00:00

Building wheels for collected packages: louvain

Building wheel for louvain (setup.py) ... done

Created wheel for louvain: filename=louvain-0.8.2-cp312-cp312-linux_x86_64.whl

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size=971831
sha256=869f56866316684d56a9dc3be5417c2429c382fccd42b50854a9bb2df7dc7aec
  Stored in directory: /root/.cache/pip/wheels/40/de/2b/bb7ed19d84727f9f299f20cd
34c42bba9c8bef7d83d2255c86
Successfully built louvain
Installing collected packages: igraph, louvain
  Attempting uninstall: igraph
    Found existing installation: igraph 1.0.0
    Uninstalling igraph-1.0.0:
      Successfully uninstalled igraph-1.0.0
ERROR: pip's dependency resolver does not currently take into account all
the packages that are installed. This behaviour is the source of the following
dependency conflicts.

leidenalg 0.11.0 requires igraph<2.0,>=1.0.0, but you have igraph 0.11.9 which
is incompatible.

Successfully installed igraph-0.11.9 louvain-0.8.2
Collecting perptpy
  Downloading perptpy-1.0.5-py3-none-any.whl.metadata (9.3 kB)
Requirement already satisfied: adjusttext in /usr/local/lib/python3.12/dist-
packages (from perptpy) (1.3.0)
Requirement already satisfied: arviz in /usr/local/lib/python3.12/dist-packages
(from perptpy) (0.22.0)
Collecting blitzgsea (from perptpy)
  Downloading blitzgsea-1.3.54.tar.gz (626 kB)
    627.0/627.0 kB
15.7 MB/s eta 0:00:00
  Preparing metadata (setup.py) ... done
Requirement already satisfied: fast-array-utils[accel,sparse] in
/usr/local/lib/python3.12/dist-packages (from perptpy) (1.3.1)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-
packages (from perptpy) (3.20.3)
Requirement already satisfied: flax in /usr/local/lib/python3.12/dist-packages
(from perptpy) (0.11.2)
Collecting funsor (from perptpy)
  Downloading funsor-0.4.5-py3-none-any.whl.metadata (7.8 kB)
Collecting mudata (from perptpy)
  Downloading mudata-0.3.3-py3-none-any.whl.metadata (10 kB)
Collecting numpyro (from perptpy)
  Downloading numpyro-0.20.0-py3-none-any.whl.metadata (37 kB)
Requirement already satisfied: openpyxl in /usr/local/lib/python3.12/dist-
packages (from perptpy) (3.1.5)
Collecting ott-jax (from perptpy)
  Downloading ott_jax-0.6.0-py3-none-any.whl.metadata (21 kB)
Collecting pubchempy (from perptpy)
  Downloading pubchempy-1.0.5-py3-none-any.whl.metadata (4.3 kB)

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Requirement already satisfied: pyarrow in /usr/local/lib/python3.12/dist-packages (from perpty) (18.1.0)

Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-packages (from perpty) (2.32.4)

Requirement already satisfied: rich in /usr/local/lib/python3.12/dist-packages (from perpty) (13.9.4)

Requirement already satisfied: scanpy in /usr/local/lib/python3.12/dist-packages (from perpty) (1.12)

Requirement already satisfied: scikit-learn>=1.4 in /usr/local/lib/python3.12/dist-packages (from perpty) (1.6.1)

Collecting scikit-misc (from perpty)

 Downloading scikit_misc-0.5.2-cp312-cp312-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl.metadata (5.0 kB)

Collecting sparsecca (from perpty)

 Downloading sparsecca-0.3.1-py3-none-any.whl.metadata (1.9 kB)

Requirement already satisfied: numpy>=1.19.5 in /usr/local/lib/python3.12/dist-packages (from scikit-learn>=1.4->perpty) (2.0.2)

Requirement already satisfied: scipy>=1.6.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn>=1.4->perpty) (1.16.3)

Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn>=1.4->perpty) (1.5.3)

Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn>=1.4->perpty) (3.6.0)

Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (from adjusttext->perpty) (3.10.0)

Requirement already satisfied: setuptools>=60.0.0 in /usr/local/lib/python3.12/dist-packages (from arviz->perpty) (75.2.0)

Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages (from arviz->perpty) (26.0)

Requirement already satisfied: pandas>=2.1.0 in /usr/local/lib/python3.12/dist-packages (from arviz->perpty) (2.2.2)

Requirement already satisfied: xarray>=2023.7.0 in /usr/local/lib/python3.12/dist-packages (from arviz->perpty) (2025.12.0)

Requirement already satisfied: h5netcdf>=1.0.2 in /usr/local/lib/python3.12/dist-packages (from arviz->perpty) (1.8.1)

Requirement already satisfied: typing-extensions>=4.1.0 in /usr/local/lib/python3.12/dist-packages (from arviz->perpty) (4.15.0)

Requirement already satisfied: xarray-einstats>=0.3 in /usr/local/lib/python3.12/dist-packages (from arviz->perpty) (0.9.1)

Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from blitzgsea->perpty) (4.67.2)

Requirement already satisfied: statsmodels in /usr/local/lib/python3.12/dist-packages (from blitzgsea->perpty) (0.14.6)

Requirement already satisfied: mpmath in /usr/local/lib/python3.12/dist-packages (from blitzgsea->perpty) (1.3.0)

Requirement already satisfied: numba>=0.57 in /usr/local/lib/python3.12/dist-packages (from fast-array-utils[accel,sparse]->perpty) (0.60.0)

Requirement already satisfied: jax>=0.6.0 in /usr/local/lib/python3.12/dist-

packages (from flax->pertpy) (0.7.2)
 Requirement already satisfied: msgpack in /usr/local/lib/python3.12/dist-packages (from flax->pertpy) (1.1.2)
 Requirement already satisfied: optax in /usr/local/lib/python3.12/dist-packages (from flax->pertpy) (0.2.6)
 Requirement already satisfied: orbax-checkpoint in /usr/local/lib/python3.12/dist-packages (from flax->pertpy) (0.11.32)
 Requirement already satisfied: tensorstore in /usr/local/lib/python3.12/dist-packages (from flax->pertpy) (0.1.80)
 Requirement already satisfied: PyYAML<=5.4.1 in /usr/local/lib/python3.12/dist-packages (from flax->pertpy) (6.0.3)
 Requirement already satisfied: treescope<=0.1.7 in /usr/local/lib/python3.12/dist-packages (from flax->pertpy) (0.1.10)
 Requirement already satisfied: markdown-it-py<=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich->pertpy) (4.0.0)
 Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich->pertpy) (2.19.2)
 Collecting makefun (from funsor->pertpy)
 Downloading makefun-1.16.0-py2.py3-none-any.whl.metadata (2.9 kB)
 Requirement already satisfied: multipledispatch in /usr/local/lib/python3.12/dist-packages (from funsor->pertpy) (1.0.0)
 Requirement already satisfied: opt-einsum<=2.3.2 in /usr/local/lib/python3.12/dist-packages (from funsor->pertpy) (3.4.0)
 Requirement already satisfied: anndata<=0.10.8 in /usr/local/lib/python3.12/dist-packages (from mudata->pertpy) (0.12.10)
 Requirement already satisfied: session-info2 in /usr/local/lib/python3.12/dist-packages (from mudata->pertpy) (0.4)
 Requirement already satisfied: jaxlib<=0.7.0 in /usr/local/lib/python3.12/dist-packages (from numpyro->pertpy) (0.7.2)
 Requirement already satisfied: et-xmlfile in /usr/local/lib/python3.12/dist-packages (from openpyxl->pertpy) (2.0.0)
 Collecting jaxopt<=0.8 (from ott-jax->pertpy)
 Downloading jaxopt-0.8.5-py3-none-any.whl.metadata (3.3 kB)
 Collecting lineax<=0.0.7 (from ott-jax->pertpy)
 Downloading lineax-0.1.0-py3-none-any.whl.metadata (18 kB)
 Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests->pertpy) (3.4.4)
 Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests->pertpy) (3.11)
 Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests->pertpy) (2.5.0)
 Requirement already satisfied: certifi<=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests->pertpy) (2026.1.4)
 Requirement already satisfied: h5py<=3.11 in /usr/local/lib/python3.12/dist-packages (from scanpy->pertpy) (3.15.1)
 Requirement already satisfied: legacy-api-wrap<=1.5 in /usr/local/lib/python3.12/dist-packages (from scanpy->pertpy) (1.5)
 Requirement already satisfied: natsort in /usr/local/lib/python3.12/dist-

packages (from scanpy->pertpy) (8.4.0)
 Requirement already satisfied: networkx>=2.8.8 in
 /usr/local/lib/python3.12/dist-packages (from scanpy->pertpy) (3.6.1)
 Requirement already satisfied: patsy in /usr/local/lib/python3.12/dist-packages
 (from scanpy->pertpy) (1.0.2)
 Requirement already satisfied: pynndescent>=0.5.13 in
 /usr/local/lib/python3.12/dist-packages (from scanpy->pertpy) (0.6.0)
 Requirement already satisfied: seaborn>=0.13.2 in
 /usr/local/lib/python3.12/dist-packages (from scanpy->pertpy) (0.13.2)
 Requirement already satisfied: umap-learn>=0.5.7 in
 /usr/local/lib/python3.12/dist-packages (from scanpy->pertpy) (0.5.11)
 Requirement already satisfied: pyomo in /usr/local/lib/python3.12/dist-packages
 (from sparsecca->pertpy) (6.9.5)
 Requirement already satisfied: array-api-compat>=1.7.1 in
 /usr/local/lib/python3.12/dist-packages (from anndata>=0.10.8->mudata->pertpy)
 (1.13.0)
 Requirement already satisfied: zarr!=3.0.*,>=2.18.7 in
 /usr/local/lib/python3.12/dist-packages (from anndata>=0.10.8->mudata->pertpy)
 (3.1.5)
 Requirement already satisfied: ml_dtypes>=0.5.0 in
 /usr/local/lib/python3.12/dist-packages (from jax>=0.6.0->flax->pertpy) (0.5.4)
 Collecting equinox>=0.11.10 (from lineax>=0.0.7->ott-jax->pertpy)
 Downloading equinox-0.13.4-py3-none-any.whl.metadata (19 kB)
 Collecting jaxtyping>=0.2.24 (from lineax>=0.0.7->ott-jax->pertpy)
 Downloading jaxtyping-0.3.7-py3-none-any.whl.metadata (7.3 kB)
 Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-
 packages (from markdown-it-py>=2.2.0->rich->pertpy) (0.1.2)
 Requirement already satisfied: contourpy>=1.0.1 in
 /usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->pertpy)
 (1.3.3)
 Requirement already satisfied: cycycler>=0.10 in /usr/local/lib/python3.12/dist-
 packages (from matplotlib->adjusttext->pertpy) (0.12.1)
 Requirement already satisfied: fonttools>=4.22.0 in
 /usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->pertpy)
 (4.61.1)
 Requirement already satisfied: kiwisolver>=1.3.1 in
 /usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->pertpy)
 (1.4.9)
 Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-
 packages (from matplotlib->adjusttext->pertpy) (11.3.0)
 Requirement already satisfied: pyparsing>=2.3.1 in
 /usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->pertpy)
 (3.3.2)
 Requirement already satisfied: python-dateutil>=2.7 in
 /usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->pertpy)
 (2.9.0.post0)
 Requirement already satisfied: llvmlite<0.44,>=0.43.0dev0 in
 /usr/local/lib/python3.12/dist-packages (from numba>=0.57->fast-array-

utils[accel,sparse]->pertpy) (0.43.0)
Requirement already satisfied: absl-py>=0.7.1 in /usr/local/lib/python3.12/dist-packages (from optax->flax->pertpy) (1.4.0)
Requirement already satisfied: chex>=0.1.87 in /usr/local/lib/python3.12/dist-packages (from optax->flax->pertpy) (0.1.90)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas>=2.1.0->arviz->pertpy) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas>=2.1.0->arviz->pertpy) (2025.3)
Requirement already satisfied: etils[epath,epy] in /usr/local/lib/python3.12/dist-packages (from orbax-checkpoint->flax->pertpy) (1.13.0)
Requirement already satisfied: nest_asyncio in /usr/local/lib/python3.12/dist-packages (from orbax-checkpoint->flax->pertpy) (1.6.0)
Requirement already satisfied: aiofiles in /usr/local/lib/python3.12/dist-packages (from orbax-checkpoint->flax->pertpy) (24.1.0)
Requirement already satisfied: protobuf in /usr/local/lib/python3.12/dist-packages (from orbax-checkpoint->flax->pertpy) (5.29.5)
Requirement already satisfied: humanize in /usr/local/lib/python3.12/dist-packages (from orbax-checkpoint->flax->pertpy) (4.15.0)
Requirement already satisfied: simplejson>=3.16.0 in /usr/local/lib/python3.12/dist-packages (from orbax-checkpoint->flax->pertpy) (3.20.2)
Requirement already satisfied: psutil in /usr/local/lib/python3.12/dist-packages (from orbax-checkpoint->flax->pertpy) (5.9.5)
Requirement already satisfied: ply in /usr/local/lib/python3.12/dist-packages (from pyomo->sparsecca->pertpy) (3.11)
Requirement already satisfied: toolz>=0.9.0 in /usr/local/lib/python3.12/dist-packages (from chex>=0.1.87->optax->flax->pertpy) (0.12.1)
Collecting wadler-lindig>=0.1.0 (from equinox>=0.11.10->lineax>=0.0.7->ott-jax->pertpy)
 Downloading wadler_lindig-0.1.7-py3-none-any.whl.metadata (17 kB)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib->adjusttext->pertpy) (1.17.0)
Requirement already satisfied: donfig>=0.8 in /usr/local/lib/python3.12/dist-packages (from zarr!=3.0.*,>=2.18.7->anndata>=0.10.8->mudata->pertpy) (0.8.1.post1)
Requirement already satisfied: google-crc32c>=1.5 in /usr/local/lib/python3.12/dist-packages (from zarr!=3.0.*,>=2.18.7->anndata>=0.10.8->mudata->pertpy) (1.8.0)
Requirement already satisfied: numcodecs>=0.14 in /usr/local/lib/python3.12/dist-packages (from zarr!=3.0.*,>=2.18.7->anndata>=0.10.8->mudata->pertpy) (0.16.5)
Requirement already satisfied: fsspec in /usr/local/lib/python3.12/dist-packages (from etils[epath,epy]->orbax-checkpoint->flax->pertpy) (2025.3.0)
Requirement already satisfied: importlib_resources in /usr/local/lib/python3.12/dist-packages (from etils[epath,epy]->orbax-checkpoint->flax->pertpy) (6.5.2)

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Requirement already satisfied: zipp in /usr/local/lib/python3.12/dist-packages
(from etils[epath,epy]->orbax-checkpoint->flax->pertpy) (3.23.0)
Downloading pertpy-1.0.5-py3-none-any.whl (225 kB)
225.2/225.2 kB
16.8 MB/s eta 0:00:00
Downloading funsor-0.4.5-py3-none-any.whl (174 kB)
174.9/174.9 kB
11.8 MB/s eta 0:00:00
Downloading mudata-0.3.3-py3-none-any.whl (43 kB)
43.4/43.4 kB
3.0 MB/s eta 0:00:00
Downloading numpyro-0.20.0-py3-none-any.whl (384 kB)
384.1/384.1 kB
18.6 MB/s eta 0:00:00
Downloading ott_jax-0.6.0-py3-none-any.whl (309 kB)
309.7/309.7 kB
14.2 MB/s eta 0:00:00
Downloading pubchempy-1.0.5-py3-none-any.whl (21 kB)
Downloading
scikit_misc-0.5.2-cp312-cp312-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl
(182 kB)
183.0/183.0 kB
11.0 MB/s eta 0:00:00
Downloading sparsecca-0.3.1-py3-none-any.whl (12 kB)
Downloading jaxopt-0.8.5-py3-none-any.whl (172 kB)
172.4/172.4 kB
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Downloading lineax-0.1.0-py3-none-any.whl (74 kB)
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4.7 MB/s eta 0:00:00
Downloading makefun-1.16.0-py2.py3-none-any.whl (22 kB)
Downloading equinox-0.13.4-py3-none-any.whl (181 kB)
181.2/181.2 kB
13.2 MB/s eta 0:00:00
Downloading jaxtyping-0.3.7-py3-none-any.whl (56 kB)
56.1/56.1 kB
3.6 MB/s eta 0:00:00
Downloading wadler_lindig-0.1.7-py3-none-any.whl (20 kB)
Building wheels for collected packages: blitzgsea
  Building wheel for blitzgsea (setup.py) ... done
  Created wheel for blitzgsea: filename=blitzgsea-1.3.54-py3-none-any.whl
size=625671
sha256=e4131b8e183b194edbf782144fa6265d982997ebcec98a25bc3ce5db45f7e54
  Stored in directory: /root/.cache/pip/wheels/e3/00/84/15131449d6f397371baa9561
4491906dbd4ca2a932194b9494
Successfully built blitzgsea
Installing collected packages: makefun, wadler-lindig, scikit-misc, pubchempy,
funsor, jaxtyping, sparsecca, numpyro, mudata, jaxopt, equinox, blitzgsea,

```



```
lineax, ott-jax, pertpy
```

```
Successfully installed blitzgsea-1.3.54 equinox-0.13.4 funsor-0.4.5 jaxopt-0.8.5  
jaxtyping-0.3.7 lineax-0.1.0 makefun-1.16.0 mudata-0.3.3 numpyro-0.20.0 ott-  
jax-0.6.0 pertpy-1.0.5 pubchempy-1.0.5 scikit-misc-0.5.2 sparsecca-0.3.1 wadler-  
lindig-0.1.7
```

```
[ ]: #Import core single cell datasets
```

```
import scanpy as sc  
import anndata as ad  
import numpy as np  
import pertpy as pt
```

```
[ ]: %matplotlib inline  
import matplotlib.pyplot as plt
```

```
[ ]: !mkdir -p PBMC  
!cd PBMC
```

```
[ ]: pbmc_mtx = pt.dt.kang_2018()
```

Output()

```
[ ]: pbmc_mtx.shape
```

```
[ ]: (24673, 15706)
```

```
[ ]: pbmc_mtx.to_df().to_csv('pbmc.csv')
```

```
[ ]: pbmc_mtx.obs.label.unique()
```

```
[ ]: ['ctrl', 'stim']  
Categories (2, object): ['ctrl', 'stim']
```

```
[ ]: pbmc_mtx.obs.rename({"label": "condition"}, axis=1, inplace=True)  
pbmc_mtx.obs["condition"].replace({"ctrl": "control", "stim": "stimulated"},  
    ↪inplace=True)
```

/tmp/ipython-input-961429376.py:2: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.

The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using

'df.method({col: value}, inplace=True)' or `df[col] = df[col].method(value)` instead, to perform the operation inplace on the original object.

```
pbmc_mtx.obs["condition"].replace({"ctrl": "control", "stim": "stimulated"},
inplace=True)
/tmp/ipython-input-961429376.py:2: FutureWarning: The behavior of Series.replace
(and DataFrame.replace) with CategoricalDtype is deprecated. In a future
version, replace will only be used for cases that preserve the categories. To
change the categories, use ser.cat.rename_categories instead.
pbmc_mtx.obs["condition"].replace({"ctrl": "control", "stim": "stimulated"},
inplace=True)
```

```
[ ]: pbmc_mtx.obs.condition.unique()
```

```
[ ]: ['control', 'stimulated']
Categories (2, object): ['control', 'stimulated']
```

```
[ ]: pbmc_mtx.obs.cell_type.value_counts()
```

```
[ ]: cell_type
CD4 T cells      11238
CD14+ Monocytes  5697
B cells          2651
NK cells         1716
CD8 T cells      1621
FCGR3A+ Monocytes 1089
Dendritic cells  529
Megakaryocytes   132
Name: count, dtype: int64
```

```
[ ]: ag_rfc = pt.tl.Augur("random_forest_classifier")
```

```
[ ]: loaded_data = ag_rfc.load(pbmc_mtx, label_col="condition",
    ↪cell_type_col="cell_type")
loaded_data
```

```
[ ]: AnnData object with n_obs × n_vars = 24673 × 15706
      obs: 'nCount_RNA', 'nFeature_RNA', 'tsne1', 'tsne2', 'label', 'cluster',
          'cell_type', 'replicate', 'nCount_SCT', 'nFeature_SCT',
          'integrated_snn_res.0.4', 'seurat_clusters', 'y_'
      var: 'name'
      obsm: 'X_pca', 'X_umap'
```

```
[ ]: v_data, v_results = ag_rfc.predict(
      loaded_data, subsample_size=50, select_variance_features=True, span=1
    )
```

```
v_results["summary_metrics"]
```

! Set smaller span value in the case of a `segmentation fault` error.
! Set larger span in case of svddc or other near singularities error.

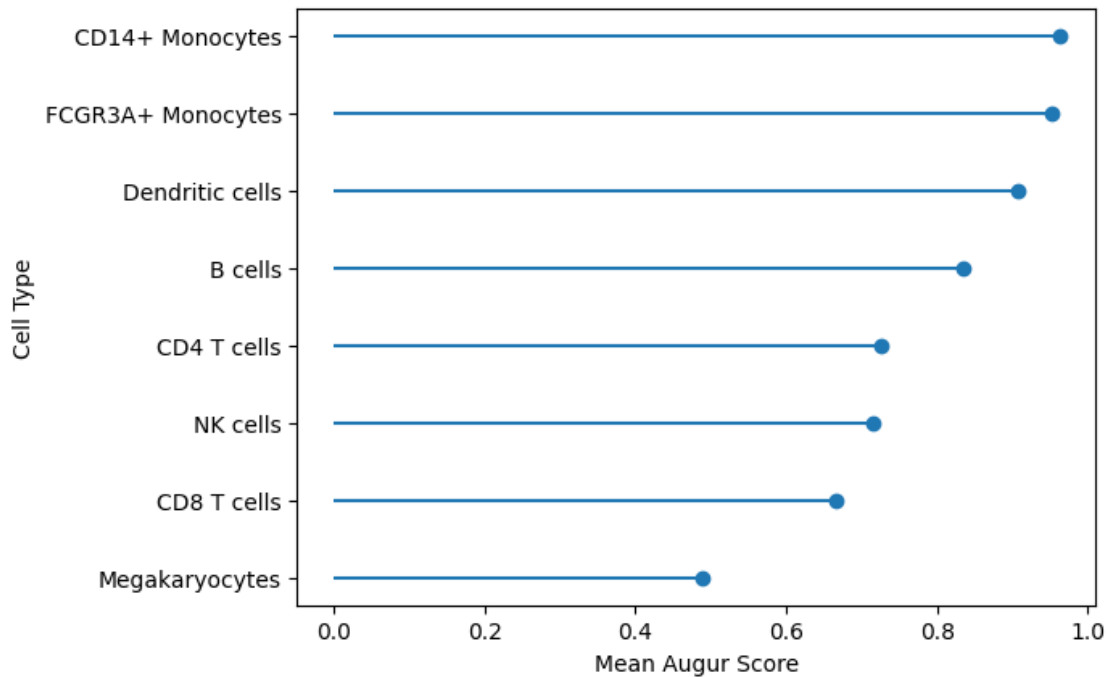
Output()

```
[ ]:
```

	CD14+ Monocytes	CD4 T cells	Dendritic cells	NK cells	\
mean_augur_score	0.962468	0.725739	0.906783	0.715087	
mean_auc	0.962468	0.725739	0.906783	0.715087	
mean_accuracy	0.889828	0.657617	0.823464	0.649346	
mean_precision	0.893741	0.679377	0.827669	0.658192	
mean_f1	0.889611	0.642497	0.824429	0.646126	
mean_recall	0.897402	0.635931	0.833922	0.661838	

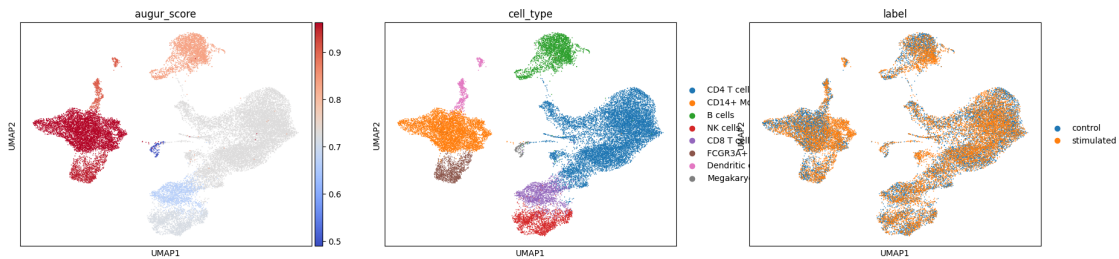
	CD8 T cells	B cells	FCGR3A+ Monocytes	Megakaryocytes
mean_augur_score	0.666344	0.834357	0.951414	0.488749
mean_auc	0.666344	0.834357	0.951414	0.488749
mean_accuracy	0.611040	0.742965	0.870582	0.504991
mean_precision	0.622147	0.784978	0.857924	0.516240
mean_f1	0.599674	0.723561	0.874997	0.412031
mean_recall	0.605490	0.691863	0.902206	0.366275

```
[ ]: lollipop = ag_rfc.plot_lollipop(v_results)
```

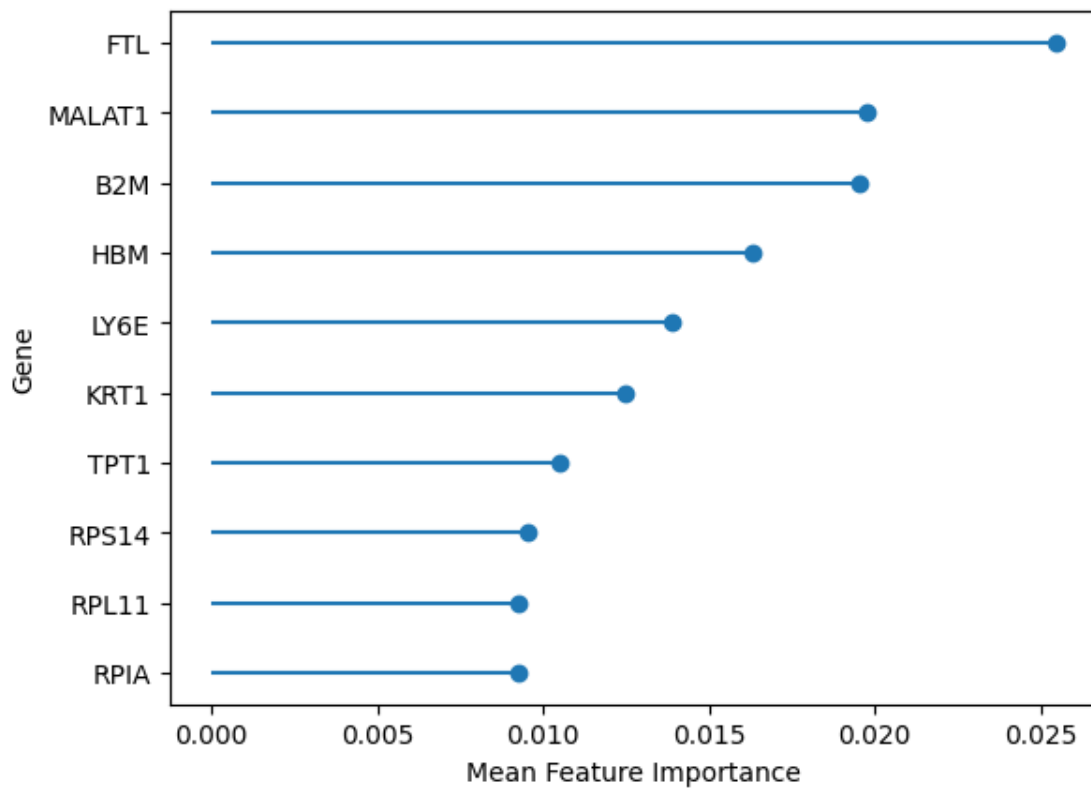


```
[ ]: sc.pp.neighbors(v_adata)
     sc.tl.umap(v_adata)
```

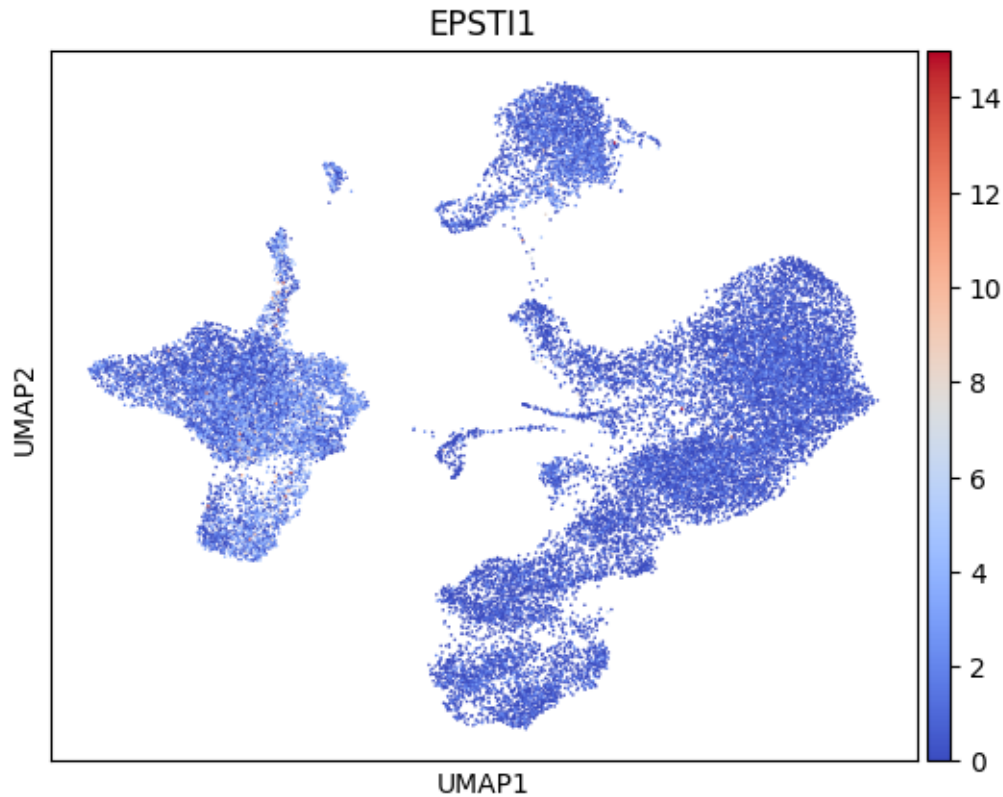
```
[ ]: sc.pl.umap(adata=v_adata, color=["augur_score", "cell_type", "label"],
               cmap='coolwarm')
```



```
[ ]: important_features = ag_rfc.plot_important_features(v_results)
```



```
[ ]: sc.pl.umap(adata=v_adata, color=["EPSTI1"], cmap='coolwarm')
```



```
[ ]: v_results.keys()
```

```
[ ]: dict_keys(['summary_metrics', 'feature_importances', 'full_results', 'CD14+ Monocytes', 'CD4 T cells', 'Dendritic cells', 'NK cells', 'CD8 T cells', 'B cells', 'FCGR3A+ Monocytes', 'Megakaryocytes'])
```

```
[ ]: v_results.get('feature_importances').head()
```

```
[ ]:
   genes  feature_importances  subsample_idx  fold  cell_type
0   TRMU           0.000000             0     0  CD14+ Monocytes
1  RAB14           0.001012             0     0  CD14+ Monocytes
2  ZNF91           0.000000             0     0  CD14+ Monocytes
3  UBE2L3          0.000131             0     0  CD14+ Monocytes
4    TBP           0.000000             0     0  CD14+ Monocytes
```

```
[ ]: feat_imp = v_results.get('feature_importances')
```

```
[ ]: feat_imp[feat_imp['cell_type'] == 'CD14+ Monocytes'].
      ↪sort_values(by='feature_importances', ascending=False).head(10)
```

```

[ ]:      genes  feature_importances  subsample_idx  fold      cell_type
129293    IRF7          0.027512          31      1  CD14+ Monocytes
165392    IRF7          0.026688          40      1  CD14+ Monocytes
162885    IRF7          0.024835          39      2  CD14+ Monocytes
93460     EPSTI1        0.021138          22      2  CD14+ Monocytes
159164    UBE2L6        0.019974          38      2  CD14+ Monocytes
189407     BST2          0.019820          46      1  CD14+ Monocytes
203430    EPSTI1        0.019333          49      2  CD14+ Monocytes
83744     BST2          0.018949          20      1  CD14+ Monocytes
2619      BST2          0.018942           0      1  CD14+ Monocytes
183638     MX2          0.018884          45      0  CD14+ Monocytes

```